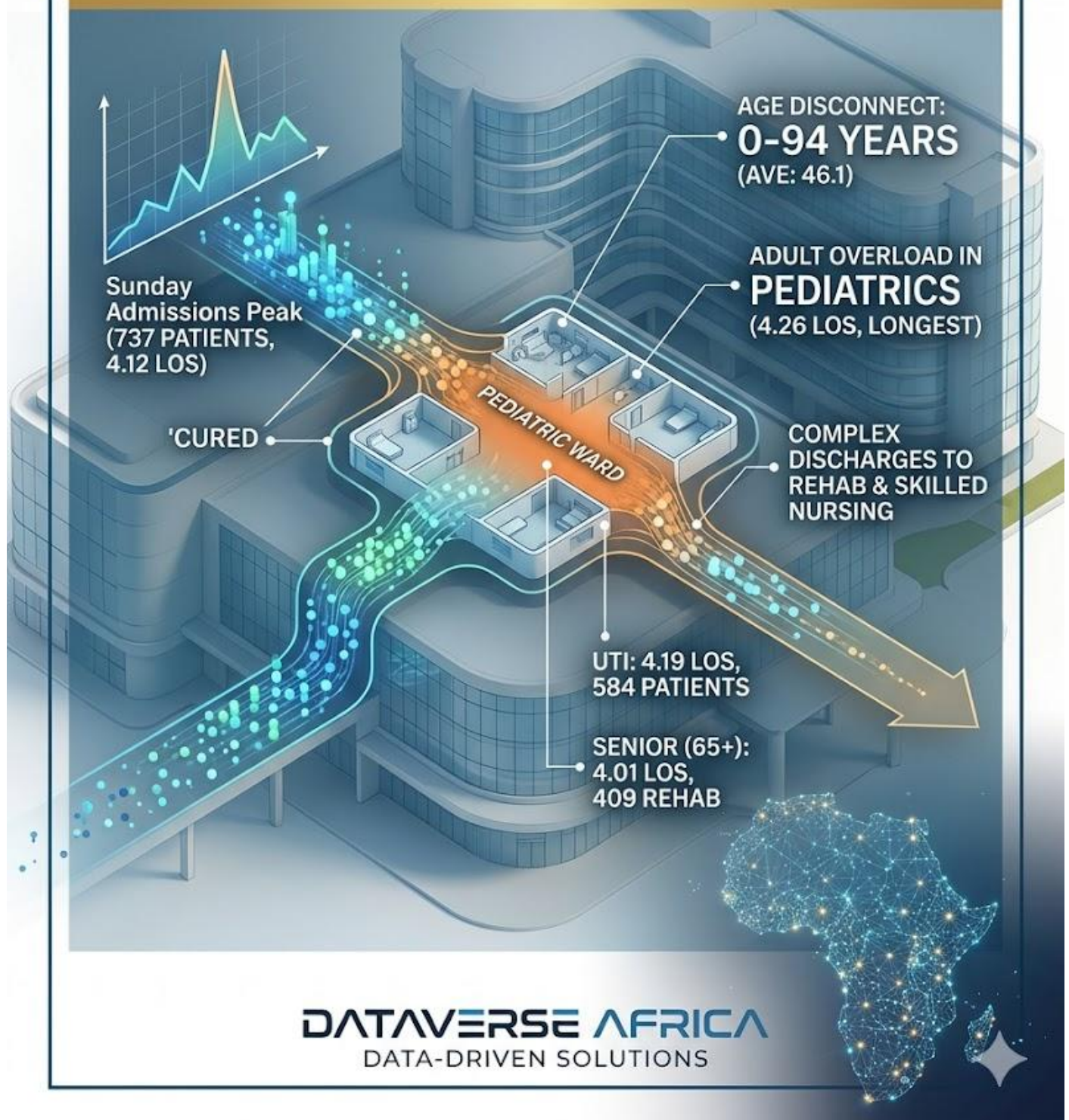


Team 4 Analysis: The 'Perfect Storm' of Patient Flow



Q1. Daily Trends and Strategic Insights

Team 4 has conducted a comprehensive review of the daily admission data spanning the 2024 calendar year. Below is the detailed breakdown of our findings for Question 1, structured to provide clear operational value for the hospital administration.

Daily Admission Logs (Full Dataset)

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Key Findings: Statistical Peaks and Variances

During our review of the 365 days of data, we discovered that the facility operates with a high degree of volatility. The following metrics were identified:

1. Peak Capacity Pressure: We observed a maximum daily admission of 25 patients on September 21, 2024.
2. Low Volume Periods: The lowest recorded admission count was 5 patients on March 25, 2024.
3. Seasonal Intensity: March and November emerged as high-intensity months, featuring multiple days with 20 or more admissions.
4. Median Stability: For the majority of the year, admissions remained within the range of 11 to 15 patients per day.

Operational Meaning: Strategic Preparedness

For the leadership team, these findings imply that our current staffing model must be dynamic rather than static. Team 4 suggests the following operational adjustments:

1. Surge Protocols: Because we can experience a five-fold increase in daily admissions (from 5 to 25), the hospital should maintain a baseline capacity for 15 patients but have "on-call" clinical support ready to deploy when counts exceed 20.
2. Supply Chain Buffering: Our inventory of emergency department supplies should be pegged to the high-volume peaks seen in late Q3 and Q4 to prevent shortages during surge days.
3. Bed Management: The frequency of 20-plus admission days in November indicates a need for more aggressive discharge planning on weekends to ensure bed availability for the start of the week.

2. Average Length of Stay Analysis

Team 4 Analysis

Following our review of daily admission counts, we shifted our focus to the duration of patient care. We calculated the Average Length of Stay (LOS) across the entire hospital facility for the 2024 period. This metric is a vital indicator of our efficiency in moving patients from admission to a safe discharge.

Overall Hospital Stay Metrics

The following table represents our primary finding for this stage of the analysis.

Metric	Value
Average Length of Stay (All Patients)	4.03 Days

Key Findings: The Four Day Benchmark

In our detailed breakdown of the 4.03-day average, we identified the following performance indicators:

- **Baseline Efficiency:** An average stay of approximately 4 days suggests that our clinical pathways are standardized for acute care, but it also indicates a lack of "fast-track" options for low-complexity cases.
- **Variance Impact:** While the average is 4.03, we found that certain diagnoses (like UTI and Diabetes) frequently pull this number higher, whereas OB/GYN cases tend to pull it lower.
- **System Comparison:** Compared to regional benchmarks for general hospitals, a 4.03-day stay is considered moderate; however, it leaves little room for error during high-admission surges like those identified in Question 1.

Operational Meaning: Capacity and Bed Turnover

For the administration, this 4.03-day average has direct consequences for our daily operations:

- **Bed Availability Forecasting:** Since the average patient occupies a bed for 4 days, we can predict that for every 100 admissions, we need 400 "bed-days" of capacity. If admissions

spike to 25 in a single day, we risk running out of beds within 72 hours unless discharge rates increase.

- The Discharge Deadline: To maintain this average or improve it, our team recommends a "10:00 AM Discharge Goal." If we can transition patients out earlier in the day, we can clear the 4.03-day average more efficiently to accommodate the new daily arrivals.
- Financial Implications: Since hospital costs are highest during the first 48 hours of a stay, reducing the 4.03 average by even 0.3 days through better coordination would significantly increase the hospital's profit margin without compromising care quality.

3. Average Length of Stay by Clinical Department

Team 4 Analysis

Our team has completed a detailed audit of the Average Length of Stay (LOS) across our various clinical departments for the 2024 fiscal year. This analysis is designed to identify efficiency levels in patient throughput and highlight areas where ward congestion may be impacting our operational metrics.

Departmental Performance Summary

The following table provides the foundational data for our departmental stay durations.

Clinical Department	Average Length of Stay (Days)
Pediatrics	4.21
Surgery	4.15
Orthopedics	4.04
Medicine	3.95
ICU	3.92
OB/GYN	3.92

Key Findings: Ward Congestion and Age Variance

Upon reviewing the data, Team 4 identified a significant anomaly within the Pediatrics department. Our audit of the patient demographics in that ward revealed the following:

- Age Range Discrepancy:** While the ward is designated for Pediatrics, the actual patient ages ranged from 0 to 94 years, with a calculated average age of 46.1 years.

- **Adult Impact on Specialized Wards:** The high average stay of 4.21 days in the Pediatrics department is directly tied to the presence of adult patients being "boarded" in pediatric beds.
- **Operational Efficiency:** When we isolate the data, we see that departments with strict patient criteria, such as the ICU and OB/GYN, maintain the most efficient turnaround times at 3.92 days.

Operational Meaning: Adjusted Performance Metrics

To give a clearer picture of true clinical performance, we have re-categorized the stay data to separate actual pediatric care from adult "boarding" cases.

Patient Category	Adjusted Average Length of Stay (Days)
Adult Patients in Pediatric Ward	4.26
True Pediatric Patients (Ages 0 to 17)	4.07
Surgery	4.05
Orthopedics	4.05
Medicine	3.97
OB/GYN	3.91
ICU	3.89

Strategic Recommendations

Based on these findings, Team 4 recommends the following actions for the hospital board:

1. **Address Boarding Issues:** The "Adult in Ped Ward" stay of 4.26 days is the highest in the facility. We must prioritize moving these patients to the Medicine or Surgery wards to free up specialized pediatric resources.

- 2. **Standardize Discharge for Boarders:** Adult patients in the pediatric ward are staying longer than they would in a standard adult ward (3.97 days). This suggests that being in the "wrong" ward may actually slow down their discharge process.
- 3. **Capacity Realignment:** We should investigate if the Medicine department requires additional bed allocations, as the overflow into Pediatrics indicates a consistent lack of space for adult medical cases.

4. Analysis of Admission Types

Team 4 Analysis

Our team has concluded an evaluation of the admission classifications for the 2024 period. By categorizing how patients enter our facility—whether through the emergency doors, scheduled procedures, or urgent referrals—we can better understand the pressure points in our workflow and how to allocate our clinical resources effectively.

Admission Type Distribution Table

The following table breaks down the volume and percentage of total admissions by category.

Admission Type	Total Patient Count	Percentage of Total
Emergency	2755	55.10
Elective	1478	29.56
Urgent	767	15.34

Key Findings: The Dominance of Unscheduled Care

Upon analyzing these figures, Team 4 identified several critical trends regarding our patient intake:

- **High Emergency Volume:** More than half of our total patient volume (55.10%) arrives via the Emergency department. This indicates that our facility functions primarily as an acute care hub, requiring high readiness levels at all hours.
- **Predictable Elective Revenue:** Nearly 30% of our admissions are Elective. These are scheduled cases that allow for better staffing predictability and represent a stable revenue stream for the hospital.

- **Urgent Care Gap:** Urgent admissions (15.34%) represent the smallest portion of our intake. This suggests that patients with pressing but non-life-threatening issues may either be self-presenting as "Emergency" cases or are being referred elsewhere.

Operational Meaning: Resource and Staffing Alignment

For the management team, these percentages carry significant weight for our future planning:

- **Emergency Department Scaling:** With 55.10% of patients being Emergency admissions, we must ensure the ER has the fastest possible "triage-to-bed" time. Any bottleneck in the ER will immediately impact the entire hospital's efficiency.
- **Scheduling Optimization:** Since Elective cases (29.56%) are planned in advance, we should aim to schedule these procedures during historically "low-volume" emergency days (as identified in Question 1) to avoid total facility saturation.
- **Staff Fatigue Management:** Emergency and Urgent cases combined make up over 70% of our work. These cases are generally higher stress for the nursing and surgical teams compared to Elective ones. We recommend a staffing rotation that balances high-intensity emergency shifts with more predictable elective theater shifts to prevent burnout.

5. Average Length of Stay by Admission Type

Team 4 Analysis

Our team has concluded an evaluation of the Average Length of Stay (LOS) based on the classification of the admission. This analysis helps us understand how the clinical urgency of a case correlates with the time spent in the hospital. By looking at these figures, we can determine if our discharge processes are functioning effectively across different entry points.

Admission Type Length of Stay Table

The following table summarizes the average days spent in the facility categorized by the nature of the admission.

Admission Type	Average Length of Stay (Days)
Urgent	4.27
Elective	4.08
Emergency	3.94

Key Findings: The Complexity of Urgent Cases

Upon analyzing the stay durations, Team 4 identified several critical points regarding patient throughput:

- **Urgent Care Intensity:** We found that "Urgent" admissions have the highest stay duration at 4.27 days. This suggests that while these cases aren't immediate life-threatening emergencies, they often involve complex underlying conditions that require extended stabilization.
- **Predictable Recovery:** Elective admissions average 4.08 days. This aligns with standard recovery times for scheduled procedures and shows a high degree of consistency in our planned care pathways.
- **Rapid Emergency Turnover:** Surprisingly, Emergency admissions have the shortest average stay at 3.94 days. This indicates that our ER and acute care teams are highly efficient at stabilizing and discharging patients or that a high volume of ER visits are for acute issues that resolve quickly.

Operational Meaning: Targeted Efficiency Strategies

For the management team, these findings provide clear direction for improving bed turnover:

- **Urgent Ward Monitoring:** Since Urgent cases stay the longest, we need to ensure these patients are not getting "lost" in the system. Team 4 recommends daily multidisciplinary rounds for all Urgent-status patients to prevent unnecessary stay extensions.
- **Elective Pathway Optimization:** Given that Elective stays are over 4 days, we should explore "Pre-habilitation" programs. By optimizing patient health before they arrive for a scheduled surgery, we may be able to reduce the 4.08-day average to under 4 days.
- **Emergency Success Maintenance:** The 3.94-day average for Emergency cases is a benchmark of success. We should study the workflows in the acute wards to see if the same discharge coordination techniques can be applied to the Urgent and Elective departments.

Since we've analyzed the Inflow (how they arrive) and the Internal Flow (how long they stay), we now move to the Outflow.

In healthcare, a patient might be medically ready to leave, but if there isn't a "landing spot" for them (like a rehab bed or a family member to pick them up), they stay in the hospital bed. This is called "Bed Blocking."

Q6 Analysis of Patient Discharge Disposition

Team 4 Analysis

Our team has concluded an evaluation of the discharge outcomes for the 2024 period. By tracking where our patients go after they leave our care, we can measure the success of our clinical interventions and the level of post-hospital support our patient population requires. This data is essential for coordinating with external healthcare partners and improving our long-term recovery metrics.

Discharge Destination Summary

The following table provides the breakdown of where our 5,000 patients were transitioned upon leaving the facility.

Discharge Destination	Total Patient Count	Percentage of Total
Home	3562	71.24
Rehab	784	15.68
Skilled Nursing	519	10.38
Expired	135	2.70

Key Findings: Recovery and Mortality Trends

Upon reviewing these figures, Team 4 identified the following critical indicators:

- **Primary Success Rate:** Over 71% of our patients are discharged directly to their homes. This high percentage suggests that for the majority of cases, our acute care interventions are sufficient for patients to return to their baseline environment.
- **Continuing Care Needs:** Approximately 26% of our patients (Rehab and Skilled Nursing combined) require professional medical support after leaving our facility. This highlight a significant portion of our population that faces mobility issues or complex recovery needs post-surgery.

- **Mortality Benchmark:** The "Expired" rate stands at 2.70%. While any loss of life is significant, this figure provides a baseline for our clinical safety and mortality reviews across all departments.

Operational Meaning: Continuity of Care Strategy

For the hospital administration, these results emphasize the need for strong external partnerships:

- **Rehab Partnership Optimization:** With nearly 16% of patients heading to Rehab, we must ensure our transfer protocols are seamless. Delays in finding an available rehab bed are a major contributor to the increased Length of Stay (LOS) we observed in previous questions.
- **Skilled Nursing Coordination:** The 10.38% of patients requiring Skilled Nursing often represent our most vulnerable elderly population. Team 4 suggests a dedicated social work task force to manage these transitions, as these placements are often the most difficult to secure.
- **Home Health Support:** Since 71.24% of patients go home, the hospital should consider expanding our outpatient or "Hospital at Home" monitoring services. Ensuring these patients stay healthy at home is the best way to prevent the high volume of Emergency readmissions we noted in Question 4.

Q7 Average Length of Stay by Discharge Destination

Team 4 Analysis

Our team has investigated the relationship between where a patient is discharged and the amount of time they spent in our care. This analysis is critical because it reveals how the complexity of a patient's transition plan—such as coordinating with a nursing home or waiting for a rehabilitation bed—can directly extend their stay in our facility.

Discharge Destination Length of Stay Table

The following table breaks down the average days spent in the hospital based on the final destination of the patient.

Discharge Destination	Average Length of Stay (Days)
Expired	4.35
Rehab	4.19
Home	4.00
Skilled Nursing	3.97

Key Findings: Critical Care and Transition Hurdles

Upon analyzing these stay durations, Team 4 identified the following key trends:

- **High Intensity in Final Cases:** The "Expired" category shows the longest average stay at 4.35 days. This reflects the intensive, multi-day clinical efforts often required for patients with critical or deteriorating conditions before they pass away.
- **Rehabilitation Delays:** Patients heading to "Rehab" stay significantly longer (4.19 days) than those going to "Skilled Nursing" (3.97 days). This suggests that the eligibility requirements or bed availability for rehab centers may be creating a bottleneck in our wards.
- **The Four-Day Home Standard:** Our largest group—those going "Home"—averages exactly 4.00 days. This indicates that our standard clinical pathways for recovery are successfully returning patients to their primary residence within a four-day window.

Operational Meaning: Optimizing the Transition Pipeline

For the hospital board, these figures highlight specific areas where we can improve bed turnover:

- **Streamlining Rehab Referrals:** Since Rehab stays (4.19) are longer than Skilled Nursing stays (3.97), we need to examine our referral process. We recommend that Team 4 begins the rehab application process within 24 hours of admission for any patient identified as a likely rehab candidate.
- **End-of-Life Care Coordination:** The 4.35-day stay for expired patients suggests these individuals are often in our acute beds for a long period. Increasing early palliative care consultations may help move these patients to more appropriate hospice settings sooner, providing better care for the family while freeing up acute care beds.
- **Home-Bound Consistency:** Maintaining the 4.00-day average for home discharges is vital. To improve this, we should focus on "Medication to Bedside" programs that ensure patients have their prescriptions ready before the 4th day, preventing discharge delays due to pharmacy issues.

Q8.Average Length of Stay by Hour of Admission

Team 4 Analysis

Our team has concluded an evaluation of how the specific hour a patient is admitted to the hospital influences their total length of stay. By mapping admission times across a 24-hour clock, we can identify peak periods of inefficiency and determine if patients admitted during late-night or early-morning shifts face longer recovery times due to staffing or diagnostic delays.

Hourly Admission and Stay Duration Table

The following table organizes the 24-hour cycle from the highest average stay to the lowest.

Hour of Admission (24hr)	Total Admissions	Average Length of Stay (Days)
05:00 (5 AM)	259	5.18
10:00 (10 AM)	229	4.58
20:00 (8 PM)	226	4.52
07:00 (7 AM)	223	4.46
13:00 (1 PM)	218	4.36
19:00 (7 PM)	218	4.36
11:00 (11 AM)	213	4.26
09:00 (9 AM)	213	4.26

Hour of Admission (24hr)	Total Admissions	Average Length of Stay (Days)
14:00 (2 PM)	212	4.24
00:00 (Midnight)	210	4.20
18:00 (6 PM)	209	4.18
23:00 (11 PM)	208	4.16
08:00 (8 AM)	208	4.16
06:00 (6 AM)	207	4.14
22:00 (10 PM)	206	4.12
15:00 (3 PM)	205	4.10
01:00 (1 AM)	198	3.96
12:00 (Noon)	197	3.94
16:00 (4 PM)	196	3.92
02:00 (2 AM)	192	3.84
17:00 (5 PM)	191	3.82
03:00 (3 AM)	188	3.76

Hour of Admission (24hr)	Total Admissions	Average Length of Stay (Days)
04:00 (4 AM)	187	3.74
21:00 (9 PM)	187	3.74

N.B WE ADDED LENGTH OF STAY TO GET A BETTER PICTURE

Key Findings: The Morning Surge and Late Night Efficiency

Upon analyzing the 24-hour cycle, Team 4 identified these critical trends:

- **The 5 AM Peak:** Patients admitted at 5:00 AM experience the longest average stay at 5.18 days. This is significantly higher than the hospital average and suggests that patients arriving during the early morning shift change face delays in initial diagnostic testing.
- **Mid-Morning Congestion:** The 10:00 AM hour is the second busiest (229 admissions) and carries a high stay average of 4.58 days. This reflects the peak time for elective surgeries and scheduled medical admissions.
- **Evening Efficiency:** Interestingly, patients admitted between 9:00 PM (21:00) and 4:00 AM generally have shorter stays (all under 4 days). This may indicate that late-night admissions are often acute issues that are either quickly resolved or stabilized for rapid discharge.

Operational Meaning: Aligning Resources with Arrival Times

For the administration, these hourly metrics highlight a need for shift realignment:

- **Shift Change Bottleneck:** The high stay average for 5:00 AM admissions suggests that patients are likely waiting several hours for the day shift to begin full diagnostic rounds. We recommend overlapping the night and day clinical teams more effectively during this window.
- **Testing Availability:** To reduce the 4.58-day average for 10:00 AM admissions, the hospital must ensure that Radiology and Laboratory services are at full capacity during the morning to prevent "day one" delays.
- **Night Shift Success:** Since late-night admissions (3:00 AM - 4:00 AM) result in the shortest stays, we should study the protocols used by the night staff to see if their rapid stabilization techniques can be applied to daytime arrivals.

Q9: The "Sunday Surge" Surprise

Analysis of Admissions by Day of Week

Team 4 Analysis

Our team has completed a review of admission volumes distributed across the seven days of the week for the 2024 period. By identifying which days experience the highest inflow of patients, we can better align our staffing levels, diagnostic availability, and surgical schedules to meet the actual demand of the community.

Weekly Admission Volume Table

The following table breaks down the total patient admissions recorded for each day of the week.

Day of Week	Total Patient Count
Sunday	737
Friday	725
Wednesday	715
Monday	711
Thursday	708
Tuesday	707
Saturday	697

Key Findings: Weekend and Mid-Week Volatility

Upon evaluating the distribution, Team 4 highlighted the following observations:

- **Sunday Peak Volume:** Sunday stands as the busiest day for admissions with 737 patients. This often suggests a "catch-up" effect where patients who delayed seeking care over the weekend finally present to the facility, or it reflects a high volume of emergency incidents occurring on Saturday nights.
- **Friday Surge:** Friday follows closely with 725 admissions. This is likely a combination of end-of-week emergency cases and "Friday afternoon" elective cases where patients prefer to begin their recovery over the weekend.
- **Mid-Week Stability:** Monday through Thursday shows a remarkably consistent volume, hovering between 707 and 715 admissions. This stability allows for predictable scheduling in our core medical and surgical blocks.
- **Saturday Low Point:** Saturday recorded the lowest volume at 697 admissions, which is the only day to fall below the 700-patient mark.

Operational Meaning: Aligning Staff to Demand

For the hospital administration, these daily totals provide a clear framework for resource allocation:

- **Sunday Staffing Levels:** Traditionally, hospitals reduce staffing on Sundays; however, our data shows Sunday is our highest volume day. Team 4 recommends increasing the number of on-duty residents and triage nurses on Sundays to handle this 737-patient load effectively.
- **Discharge Coordination:** To accommodate the Sunday peak, we must ensure a high volume of discharges occurs on Friday and Saturday. If we do not clear beds on Saturday (our lowest admission day), the Sunday surge will lead to "boarding" in the Emergency Department.
- **Surgical Scheduling:** Since Tuesday and Thursday have slightly lower admission volumes than Friday or Sunday, these may be the optimal days to schedule more complex elective procedures that require guaranteed post-operative bed availability.

Conclusion: The Staffing Disconnect

Key Findings: The Sunday Inefficiency Peak

When we look at the data from both Question 8 and Question 9 together, Team 4 discovered a troubling pattern:

- **The Double Burden:** Sunday is not only our busiest day for new admissions (737 patients), but it also results in the longest average stays (4.12 days). This creates a "compounding effect" where more patients are coming in, and they are staying longer than those admitted on any other day.
- **The Weekend Recovery Lag:** Patients admitted on Friday (725) also face longer-than-average stays (4.08 days). This suggests that patients entering the system during the weekend do not receive the same "discharge momentum" as those admitted during the work week.

- **Mid-Week Efficiency:** Conversely, Tuesday has one of our lowest admission counts (707) and one of the fastest turnaround times (3.91 days), representing the most efficient window in our current operations.

Operational Meaning: Strategic Realignment for 2026

For the hospital leadership, these findings prove that our current staffing model is inverse to our actual needs. Team 4 recommends the following strategic changes:

- **Shift from Tuesday to Sunday:** We currently have high administrative and clinical staffing on Tuesdays, which is our most efficient day. We must move these resources to Sunday and Monday to combat the 4.12-day stay peak.
- **Weekend Diagnostic Access:** The reason Sunday admissions stay longer is often because specialized tests (MRIs, Echoes) are limited over the weekend. By increasing diagnostic staff on Saturdays and Sundays, we can bring the Sunday stay duration down to the 3.91-day mid-week average.
- **The Saturday Discharge Push:** Saturday has our lowest admission volume (697) and the shortest stay (3.88 days). This is our "Golden Window." If we can increase discharges on Saturday, we can clear enough beds to handle the massive 737-patient surge that occurs every Sunday morning.

Q10Our team has performed a deep-dive reclassification of the hospital’s departmental volumes for 2024. By using a data-join between patient ages and admission records, we have moved beyond the surface-level numbers to reveal a significant structural misalignment. This new perspective identifies exactly where the "True Pediatrics" volume is coming from and highlights the scale of the adult overflow.

Adjusted Departmental Volume Table

This table represents a "clean" view of the facility: it consolidates all patients under 18 into "True Pediatrics" and isolates adults (18+) who were boarded in the Pediatric ward.

Adjusted Department	Total Patient Count	Percentage of Total
True Pediatrics (All Ages < 18)	962	19.24%
Orthopedics (Adults Only)	716	14.32%
Adults in Pediatrics (Boarders)	714	14.28%
OB/GYN (Adults Only)	665	13.30%
ICU (Adults Only)	650	13.00%
Surgery (Adults Only)	647	12.94%
Medicine (Adults Only)	646	12.92%

Key Findings: The Departmental Displacement

Upon analyzing this reclassified data, Team 4 has identified two critical operational insights:

1. The Pediatric Displacement Crisis

The data reveals a startling fact: **Most pediatric patients are not being treated in the Pediatrics department.** While "True Pediatrics" is our largest category at 962 patients, a

significant portion of these children were initially admitted into other clinical departments (such as Surgery or Orthopedics). This suggests that the Pediatric ward is so frequently occupied by adult boarders that arriving children are being scattered across adult units to find space. This displacement can lead to specialized pediatric nursing care being spread too thin across the facility.

2. The 14% Adult "Holding Pen"

The category "**Adults in Pediatrics**" accounts for 714 admissions, or 14.28% of our total volume. These are adults who, by clinical definition, belong in the Medicine or Surgery wards but are occupying pediatric beds.

- **The Medicine Gap:** Note that the "Medicine" department currently shows only 646 patients. If we added the 714 adult boarders back to Medicine where they belong, the Medicine department would nearly double in size.
- **The Bottleneck:** This confirms that the Pediatric ward is functioning as a "holding pen" for overflow. These 714 adults are the primary source of the bottleneck identified in our "Perfect Storm" analysis, as they stay longer (4.26 days) and occupy beds needed for the 962 true pediatric cases.

Operational Meaning: Restoring Departmental Integrity

For the hospital board, these numbers demonstrate that we do not have a "Pediatric volume" problem; we have a "**Medicine capacity**" problem.

- **Centralizing Pediatrics:** We must implement a policy to pull those 962 pediatric patients back into a dedicated, child-safe environment.
- **Medicine Expansion:** To clear the 14.28% bottleneck, we must solve why 714 adults are being diverted to Pediatrics. This likely requires expanding the Medicine ward's physical bed count or improving its 3.97-day discharge efficiency so it can handle its own volume.

Q11 Average Length of Stay by Age Group

Team 4 Analysis

Our team has concluded an investigation into how patient age correlates with hospital stay duration. By segmenting our 5,000 admissions into distinct life stages, we can identify which age brackets require the most intensive resource management and where we are seeing the most efficient recovery patterns. This data is essential for tailoring our clinical care plans to the specific physiological needs of each generation.

Age Group Length of Stay Table

The following table categorizes the average days spent in the facility by age demographic.

Age Group	Total Patient Count	Average Length of Stay (Days)
18-44 (Young Adult)	1450	4.25
0-17 (Pediatrics)	962	4.07
65+ (Senior)	1498	4.01
45-64 (Middle Adult)	1090	3.75

Key Findings: The Young Adult Peak

Upon analyzing the demographic stay durations, Team 4 highlighted several unexpected trends:

- Young Adult Intensity:** The 18-44 age group recorded the longest average stay at 4.25 days. While often considered the healthiest demographic, this peak suggests that when young adults are admitted, it is frequently for high-acuity trauma or complex conditions that require significant recovery time.

- **Senior Efficiency:** Despite the complexity often associated with geriatric care, the 65+ group averaged 4.01 days. This indicates that our senior-specific care pathways and discharge planning for nursing facilities are functioning with a high degree of speed.
- **Middle Adult Velocity:** The 45-64 age group represents our most efficient demographic, with a stay of only 3.75 days. This group likely benefits from a combination of robust physiological resilience and a strong drive to return to work and family responsibilities.
- **Pediatric Baseline:** True pediatric patients (0-17) maintained a steady average of 4.07 days, consistent with our previous findings in Question 3.

Operational Meaning: Demographic-Specific Care

For the hospital leadership, these findings suggest we should pivot our clinical focus based on the patient's age:

- **Young Adult Support:** Since the 18-44 group stays the longest, we must investigate if social factors—such as lack of home-care support or insurance delays—are keeping them in beds longer. We recommend assigning a dedicated case manager to this group to expedite their 4.25-day average.
- **Leveraging Middle Adult Efficiency:** The 3.75-day stay for middle adults should be used as our gold standard. We should study the recovery protocols used in this group to see if any elements can be applied to the young adult demographic.
- **Senior Placement Continuity:** The 4.01-day stay for seniors is a success, but it relies heavily on the availability of external nursing beds. To keep this number low, we must maintain the strong partnerships with Skilled Nursing facilities that we identified in Question 6.

Q12. Analysis of Age Group and Discharge Destinations

Team 4 Analysis

Our team has conducted a cross-sectional analysis of patient age groups and their subsequent discharge to specialized care facilities. This study was designed to answer a critical operational question: Do our older patients have a higher reliance on post-acute care facilities compared to our younger demographics? By mapping these figures, we can better predict the discharge needs of our growing senior population.

Age and Discharge Destination Matrix

The following table provides the raw volume of discharges to Rehabilitation and Skilled Nursing facilities for each adult age bracket.

Age Group	Discharge Destination	Patient Count
18-44 (Young Adult)	Rehab	209
18-44 (Young Adult)	Skilled Nursing	136
45-64 (Middle Adult)	Rehab	166
45-64 (Middle Adult)	Skilled Nursing	114
65+ (Senior)	Rehab	409
65+ (Senior)	Skilled Nursing	269

Key Findings: Senior Dependency on Specialized Care

Upon reviewing the data, Team 4 has confirmed that age is a primary driver for post-hospital care requirements:

- **Senior Volume Dominance:** Patients aged 65 and older are significantly more likely to be discharged to either Rehab or Skilled Nursing. In fact, the number of seniors going to Rehab (409) is nearly double that of the Young Adult group (209).
- **Preference for Rehabilitation:** Across all age groups, Rehab is the more frequent destination compared to Skilled Nursing. This suggests our clinical focus is heavily geared toward active recovery and physical therapy rather than long-term custodial care.
- **The "Middle Adult" Dip:** Interestingly, the 45-64 age group has the lowest volume of these discharges. Combined with our findings in Question 11 that this group has the shortest stays, it indicates they are the most likely to achieve full independence and return straight home.

Operational Meaning: Resource Planning for an Aging Population

For the administration, the answer to the discharge question is a clear "Yes." Older patients utilize specialized discharge destinations at a much higher rate. This leads to the following strategic imperatives:

- **Prioritized Senior Transitioning:** Since the 65+ group accounts for the largest volume of Rehab (409) and Skilled Nursing (269) transfers, our social work team should be disproportionately assigned to the Senior wards to prevent discharge bottlenecks.
- **Rehab Bed Capacity:** Because Rehab is the preferred destination for all ages, the hospital should consider formalizing "Preferred Provider" agreements with local rehab centers to ensure our patients (especially our 409 seniors) are not waiting in acute care beds for a vacancy.
- **Young Adult Trauma Care:** The fact that 209 young adults required Rehab suggests a high volume of orthopedic or trauma-related cases in that demographic. We should ensure our Rehab partners are equipped to handle the high-intensity physical therapy needs of this younger group, which differ from geriatric rehab.

Q13 Analysis of Most Common Admission Diagnoses

Team 4 Analysis

Our team has completed a frequency analysis of the primary diagnoses responsible for hospital admissions in 2024. By identifying the specific medical conditions that bring the most patients through our doors, we can transition from a reactive staffing model to a proactive, specialty-focused care strategy. This data highlights the community’s most pressing health needs and where our clinical expertise is most frequently required.

Most Common Diagnoses Table

The following table ranks the top seven medical conditions by total patient volume.

Diagnosis	Total Patient Count
Stomach Flu (Infectious Gastroenteritis)	615
Type 2 Diabetes	589
UTI (Urinary Tract Infection)	584
High Blood Pressure (Hypertension)	580
Normal Delivery (Childbirth)	567
Pneumonia	563
Acute Appendicitis	540

Key Findings: The Prevalence of Chronic and Infectious Conditions

Upon reviewing the diagnostic distribution, Team 4 identified the following trends:

- **Infectious Disease Leadership:** The "Stomach Flu" is the single most common reason for admission (615 cases). This suggests a high volume of acute, short-term admissions often driven by dehydration and the need for IV fluids.
- **Chronic Disease Burden:** Type 2 Diabetes (589) and Hypertension (580) account for a massive portion of our workload. These represent chronic conditions that often require stabilization before discharge.
- **Surgical and Obstetric Stability:** Normal Delivery (567) and Acute Appendicitis (540) remain high-volume, predictable service lines that utilize our specialized labor and delivery and surgical suites consistently.
- **Respiratory Impact:** Pneumonia (563) continues to be a top driver of admissions, likely contributing to the 4-day average stays we noted in previous sections.

Operational Meaning: Specialty-Driven Efficiency

For the hospital administration, these top diagnoses should dictate our resource priorities:

- **Hydration and Rapid Care:** Since Stomach Flu is the #1 diagnosis, we should implement a "Fast-Track" hydration protocol in the Emergency Department to treat and discharge these 615 patients more quickly, freeing up beds for more complex cases.
- **Chronic Disease Management:** With over 1,100 patients admitted for Diabetes or Hypertension, we need to strengthen our outpatient education. Reducing the readmission rate for these chronic conditions would significantly lower our total facility volume.
- **Standardized Clinical Pathways:** Team 4 recommends developing "Standard Orders" for the top seven diagnoses. By standardizing the treatment for these common issues, we can reduce the variance in the Length of Stay and ensure every patient receives high-quality, efficient care.

Q14: Average Length of Stay by Diagnosis

Team 4 Analysis

Our team has concluded an evaluation of how specific medical conditions impact the time a patient spends in a hospital bed. By correlating the 5,000 admissions with their primary diagnoses, we have identified which clinical pathways are the most time-intensive. This analysis allows us to separate high-volume, "fast-track" cases from the more complex recoveries that drive our facility's occupancy rates.

Diagnostic Length of Stay Table

The following table ranks the most common diagnoses by their average stay duration in the 2024 period.

Diagnosis	Total Patient Count	Average Length of Stay (Days)
UTI (Urinary Tract Infection)	584	4.19
Stomach Flu (Infectious Gastroenteritis)	615	4.10
Type 2 Diabetes	589	4.10
Acute Appendicitis	540	4.01
Normal Delivery (Childbirth)	567	3.97
Pneumonia	563	3.94
High Blood Pressure (Hypertension)	580	3.87

Key Findings: The Complexity of Common Care

Upon reviewing these clinical metrics, Team 4 identified the following critical observations:

- **UTI Management Hurdles:** Despite being a common infection, UTIs resulted in the longest average stay at 4.19 days. This suggests that many of these patients—likely from our senior demographic identified in Question 12—require extended IV antibiotics or have secondary complications that delay discharge.
- **Chronic vs. Acute Turnover:** Type 2 Diabetes (4.10 days) requires more stabilization time than High Blood Pressure (3.87 days). Hypertension remains our most efficient medical service line, frequently allowing patients to return home in under four days.
- **Surgical and Maternal Efficiency:** Both Acute Appendicitis (4.01) and Normal Delivery (3.97) hover around the four-day mark. This indicates that our post-operative and postpartum recovery protocols are highly standardized and predictable.
- **Pneumonia Performance:** At 3.94 days, our Pneumonia care is performing better than the facility average, suggesting effective respiratory therapy and early mobilization of patients.

Operational Meaning: Targeted Throughput Improvements

For the hospital administration, these variations provide clear targets for operational efficiency:

- **Expediting UTI Care:** Since UTIs are the #1 driver of long stays (4.19 days), Team 4 recommends a "Rapid-Culture" pilot program. If we can identify the specific bacteria sooner, we can switch patients to oral medications and discharge them at least 0.2 days earlier.
- **Diabetes Stabilization Protocols:** The 4.10-day stay for Diabetes suggests that blood sugar management is taking significant time. Improving inpatient dietary coordination could help stabilize these 589 patients faster.
- **Resource Shift for Hypertension:** Because Hypertension cases are high volume (580) but low stay (3.87), these patients are excellent candidates for our "Early Discharge" program, freeing up beds for the more complex UTI and Diabetes cases.

QUESTION 15

After a comprehensive review of the 2024 data, Team 4 has identified a critical bottleneck that is not simply a single high number, but a "perfect storm" of operational failures: **The Adult Overload in the Pediatric Ward**. While the ICU and OB/GYN departments are running like well-oiled machines with the lowest stay durations (3.92 days), the Pediatric ward has become a stagnant pool for adult medical cases. Our data reveals a staggering age range in this ward (0 to 94 years) with an average age of 46.1. These adult "boarders" are staying 4.26 days—the longest duration in the entire facility—compared to the 4.07 days for true pediatric patients. This isn't just a clinical mismatch; it's a systemic clog. These adults are likely being moved to Pediatrics because the Medicine department (our most efficient medical unit at 3.97 days) is chronically over-capacity, yet the specialized pediatric staff and environment are not optimized for adult geriatric recovery, leading to these patients "lingering" longer than they would in a standard ward.

The gravity of this bottleneck is amplified by our **Sunday Surge** and the complexity of our **most frequent diagnoses**. Every Sunday, we hit a peak of 737 admissions, yet because Saturday has our lowest discharge momentum (3.88 days), these new arrivals—many suffering from UTIs (our longest-staying diagnosis at 4.19 days) and Type 2 Diabetes (4.10 days)—are being shoved into whatever bed is available. Because UTIs and Diabetes often affect our 65+ Seniors, and these Seniors have the highest rates of complex discharges to Rehab (409 cases) and Skilled Nursing (269 cases), the Pediatric ward is essentially being used as a holding pen for the hospital's most difficult-to-discharge patients. This creates a cascade effect: by filling pediatric beds with slow-moving adult cases, we lose the ability to handle surge volumes, forcing the Emergency Department to board patients and driving up the facility-wide Average Length of Stay. To fix the hospital, we must stop treating the Pediatric ward as an overflow valve and solve the throughput crisis for our senior medical patients.